

NEXT GEN EMPLOYABILITY PROGRAM

CREATING A
FUTURE-READY
WORKFORCE

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CAPSTONE PROJECT SHOWCASE

Project Title: Personal Finance Manager with MERN
Technology (P4)

Abstract | Problem Statement | Project Overview | Proposed Solution |
Technology Used | Modelling & Results | Conclusion | Q&A

Creating A Future-ready Workforce

Abstract

1

Users can create, view, edit, and delete expenses with real-time updates, ensuring accurate financial tracking

2

Users can log in or sign up to access personalized features like adding favorites, rating recipes, and commenting.

3

Any modifications to expenses are instantly reflected in reports, leveraging WebSockets or state management for dynamic updates.

4

Built using React.js, Node.js, Express.js, and MongoDB, it offers a secure, interactive, and scalable

Problem Statement

Problem Statement – Personal Finance Manager using MERN Stack

Managing personal finances effectively is a challenge for many individuals due to the lack of a centralized system for tracking expenses. Existing solutions may lack real-time updates, intuitive interfaces, or robust security features, making financial management cumbersome. Users often struggle to monitor their spending, categorize transactions, and generate insightful reports.

This project aims to develop a **Personal Finance Manager** using the **MERN stack**, providing users with a seamless platform to **track, manage, and analyze** their expenses. The system will ensure **real-time updates**, secure authentication, and dynamic reporting, helping users make informed financial decisions effortlessly.



Project Overview

Project Overview – Personal Finance Manager (MERN Stack)

- 1.Expense Management** – Users can add, edit, delete, and categorize expenses, ensuring an organized financial record.
- 2.User Authentication** – Secure login and registration using JWT authentication and bcrypt for password encryption.
- 3.Real-Time Updates & Reports** – Any changes in expenses reflect instantly, providing users with up-to-date financial insights through interactive charts and dashboards.
- 4.Full-Stack MERN Implementation** – Built using MongoDB (database), Express.js (backend API), React.js (frontend UI), and Node.js (server-side logic) for a seamless and scalable application.



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Proposed Solution

Personal Finance Manager (MERN Stack)

1. Secure User Authentication

- Users must log in or sign up to access their personal finance dashboard.
- JWT (JSON Web Tokens) will be used for secure authentication and session management.

2. Expense Management

- Users can add, edit, delete, and categorize expenses.
- Each transaction will be linked to the user's profile and stored in a database.

3. Real-Time Financial Insights

- Users can view dynamic reports and charts summarizing their spending patterns.
- Instant updates ensure that any changes reflect immediately in financial reports.

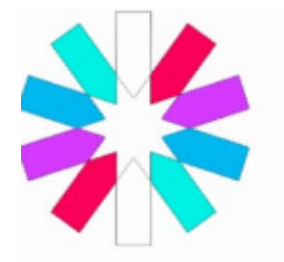
4. Role-Based Access Control

- Only authenticated users can manage their own expenses.
- Admin users (if applicable) can access and manage financial data for reporting purposes.

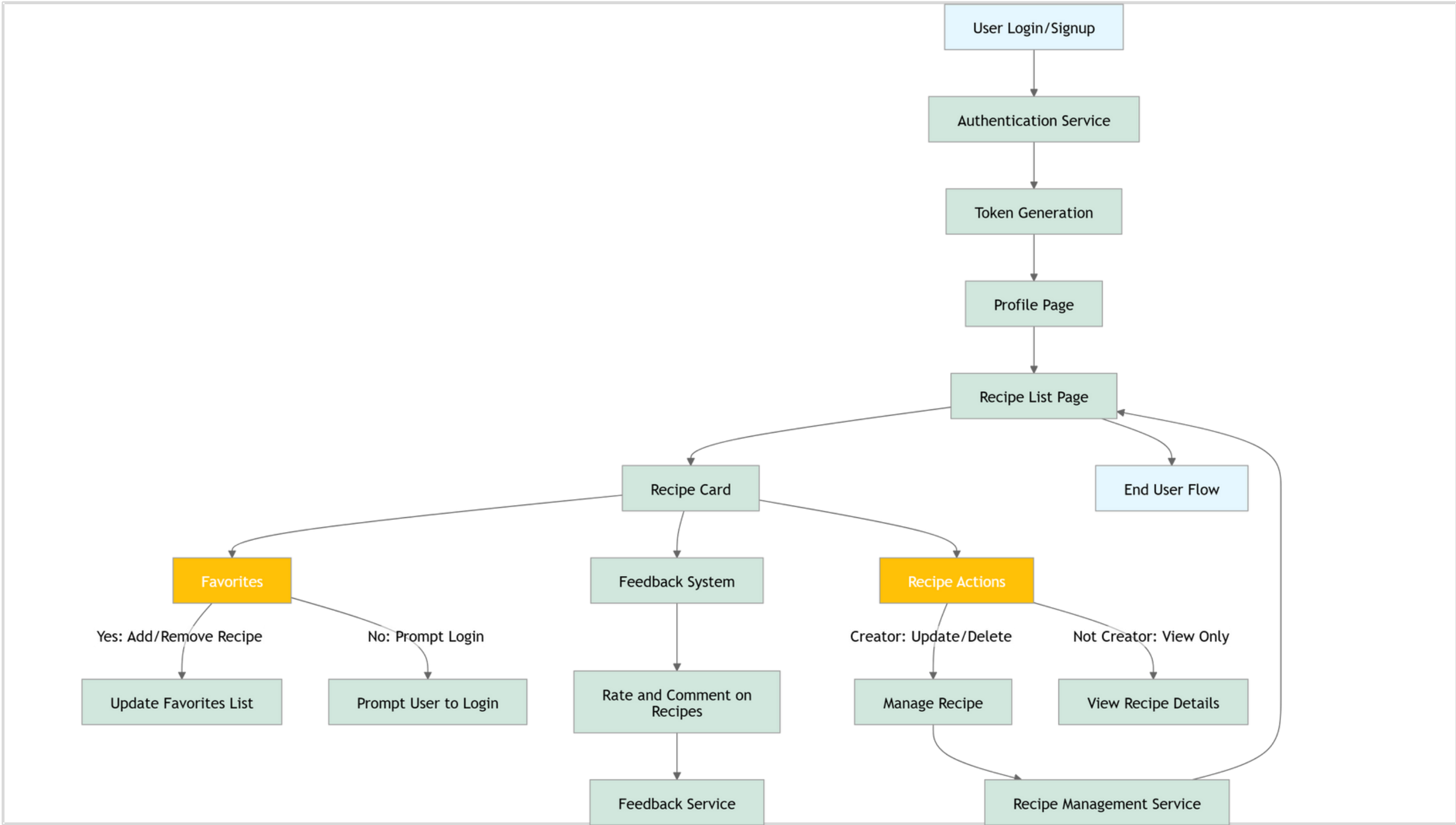
Let me know if you need modifications! 🚀

Technology used

- Frontend: HTML, CSS, JavaScript, React.js
- Backend: Node.js, Express.js
- Database: MongoDB
- Authentication: JWT (JSON Web Tokens)
- Hosting & Deployment: Render
(for backend and frontend deployment)

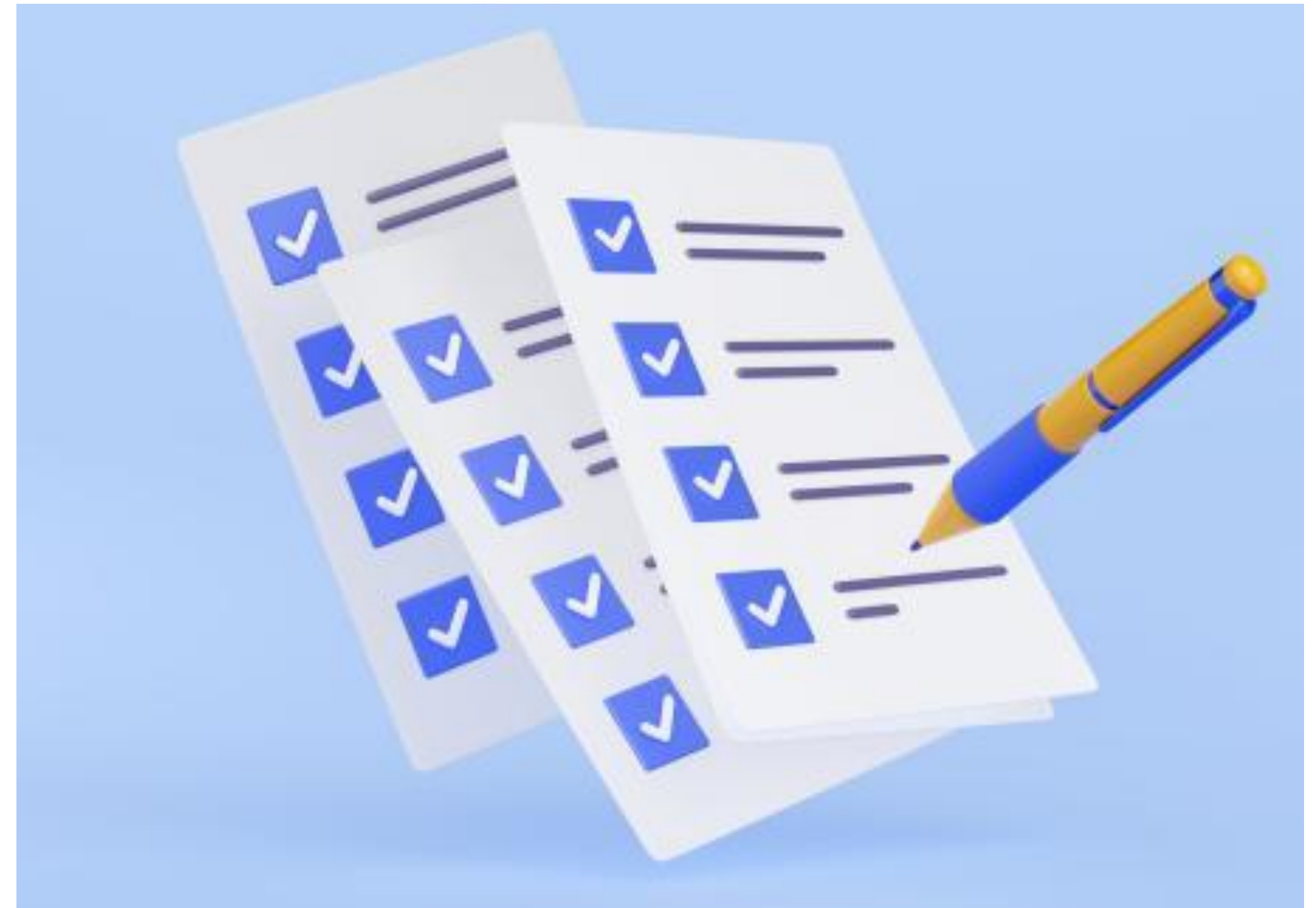


Modelling & Result : Architecture Diagram



Conclusion

The **Personal Finance Manager** using the **MERN stack** provides a seamless and efficient way for users to **track, manage, and analyze** their expenses. With **secure authentication, real-time updates, and interactive financial reports**, users can make informed financial decisions. The platform ensures **data privacy and accessibility**, allowing users to monitor their spending habits effortlessly. By integrating **MongoDB, Express.js, React.js, and Node.js**, the system delivers a scalable and responsive user experience. Overall, this project enhances financial management by offering a **secure, user-friendly, and insightful** solution. 🚀





Thank you!

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